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Subject: Mathematics **CBSE CLASS 10 – PERIODIC ASSESSMENT 07** **Time:** 60 Minutes
Class: Class X (CBSE) **Chapter 11: Areas Related to Circles** **MM: 30**

Document Type: Official Step-by-Step Marking Scheme **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper physical units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: MULTIPLE CHOICE

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Q1. A circular garden of radius 21 m is surrounded by a path 3.5 m wide. The area of the path is: $\left(\text{Use } \pi = \frac{22}{7}\right)$ [1]

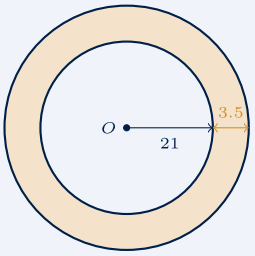
(A) 500.5 m^2 (B) 462 m^2 (C) 539 m^2 (D) 484 m^2

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Determine Inner and Outer Radii [0.5 Mark]

Inner radius $r = 21$ m. Outer radius $R = 21 + 3.5 = 24.5$ m.

$$\text{Area of path} = \pi(R^2 - r^2) = \pi(R - r)(R + r)$$



Step 2: Factoring and Arithmetic Calculation [0.5 Mark]

$$\text{Area} = \frac{22}{7} \times 3.5 \times (24.5 + 21) = 11 \times 45.5 = 500.5 \text{ m}^2$$

Correct Option: (A) 500.5 m^2

Area of the path = **500.5** m^2

Q2. The perimeter of a quadrant of a circle of radius 14 cm is: $\left(\text{Use } \pi = \frac{22}{7}\right)$ [1]

(A) 22 cm (B) 50 cm (C) 36 cm (D) 44 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Formula for Perimeter of a Quadrant [0.5 Mark]

$$\text{Perimeter} = \frac{1}{4}(2\pi r) + 2r = \frac{\pi r}{2} + 2r$$

Step 2: Substitution and Simplification [0.5 Mark]

$$\text{Perimeter} = \frac{1}{2} \times \frac{22}{7} \times 14 + 2(14) = 22 + 28 = 50 \text{ cm}$$

Correct Option: (B) 50 cm

Perimeter of quadrant = **50** cm

Q3. If the perimeter of an equilateral triangle is 18 cm, the radius of its circumscribed circle is: [1]

(A) $2\sqrt{3}$ cm (B) $\sqrt{3}$ cm (C) $3\sqrt{3}$ cm (D) 6 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Side of Equilateral Triangle [0.5 Mark]

$$\text{Side } a = \frac{\text{Perimeter}}{3} = \frac{18}{3} = 6 \text{ cm}$$

Step 2: Circumradius R [0.5 Mark]

$$R = \frac{a}{\sqrt{3}} = \frac{6}{\sqrt{3}} = \frac{6\sqrt{3}}{3} = 2\sqrt{3} \text{ cm}$$

Correct Option: (A) $2\sqrt{3}$ cm

$$\boxed{\text{Circumradius } R = \mathbf{2\sqrt{3}} \text{ cm}}$$

Q4. If a sector of a circle of radius 6 cm has perimeter 20 cm, then its area is: [1]

(A) 24 cm^2 (B) 48 cm^2 (C) 12 cm^2 (D) 36 cm^2

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Find Arc Length l [0.5 Mark]

$$\text{Perimeter} = l + 2r = 20 \implies l + 2(6) = 20 \implies l = 20 - 12 = 8 \text{ cm}$$

Step 2: Area of the Sector [0.5 Mark]

$$\text{Area} = \frac{1}{2}lr = \frac{1}{2} \times 8 \times 6 = 24 \text{ cm}^2$$

Correct Option: (A) 24 cm^2

$$\boxed{\text{Area of sector} = \mathbf{24} \text{ cm}^2}$$

Q5. If the difference between the circumference and the radius of a circle is 37 cm, then the diameter of the circle is: $\left(\text{Use } \pi = \frac{22}{7}\right)$ [1]

(A) 7 cm (B) 14 cm (C) 21 cm (D) 28 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Solve for Radius r [0.5 Mark]

$$2\pi r - r = 37 \implies r(2\pi - 1) = 37 \implies r\left(2 \times \frac{22}{7} - 1\right) = 37$$

$$r\left(\frac{44 - 7}{7}\right) = 37 \implies r \times \frac{37}{7} = 37 \implies r = 7 \text{ cm}$$

Step 2: Calculate Diameter [0.5 Mark]

$$\text{Diameter } d = 2r = 2 \times 7 = 14 \text{ cm}$$

Correct Option: (B) 14 cm

$$\boxed{\text{Diameter of circle} = \mathbf{14} \text{ cm}}$$

Q6. **Assertion (A):** Three sectors of central angles $35°, 65°$, and $80°$ cut from a circle of radius r together have an area equal to half the area of the circle.
Reason (R): The sum of the central angles is $35° + 65° + 80° = 180°$, which is half of the total central angle $360°$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Sum of Sector Angles [0.5 Mark]

$$\theta_{\text{total}} = 35° + 65° + 80° = 180° = \frac{360°}{2}$$

Reason (R) is **True**.

Step 2: Combined Sector Area [0.5 Mark]

$$\text{Total Area} = \frac{180°}{360°} \times \pi r^2 = \frac{1}{2}\pi r^2 = \frac{1}{2}(\text{Area of circle})$$

Assertion (A) is **True**, and Reason (R) is the exact justification.

Correct Option: **(A) Both (A) and (R) are true and (R) is the correct explanation of (A).**

Correct Option: (**A**)

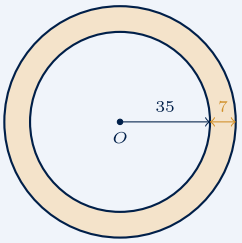
Q7. The inner circumference of a circular track is 220 m. The track is 7 m wide everywhere. Find the outer radius and outer circumference of the track. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Compute Inner and Outer Radii [1 Mark]

$$2\pi r = 220 \implies 2 \times \frac{22}{7} \times r = 220 \implies r = \frac{220 \times 7}{44} = 35 \text{ m}$$

$$\text{Outer radius } R = r + w = 35 + 7 = 42 \text{ m}$$



Step 2: Calculate Outer Circumference [1 Mark]

$$\text{Outer Circumference} = 2\pi R = 2 \times \frac{22}{7} \times 42 = 44 \times 6 = 264 \text{ m}$$

$$\boxed{\text{Outer Radius } R = \mathbf{42} \text{ m}, \quad \text{Outer Circumference} = \mathbf{264} \text{ m}}$$

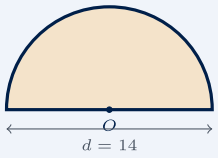
Q8. If the perimeter of a semicircular protractor is 36 cm, find its diameter. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Formula and Radius Calculation [1 Mark]

$$\text{Perimeter} = \pi r + 2r = r(\pi + 2) = 36 \text{ cm}$$

$$r\left(\frac{22}{7} + 2\right) = 36 \implies r\left(\frac{36}{7}\right) = 36 \implies r = 7 \text{ cm}$$



Step 2: Determine Diameter [1 Mark]

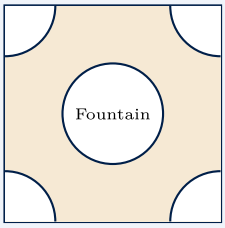
$$\text{Diameter } d = 2r = 2 \times 7 = 14 \text{ cm}$$

$$\boxed{\text{Diameter of protractor} = \mathbf{14} \text{ cm}}$$

Q9. A square park has side 50 m. At each of its four corners, a quadrant flower bed of radius 7 m is developed and a circular fountain of diameter 14 m is built at the centre. Find the remaining area of the park. $\left(\text{Use } \pi = \frac{22}{7}\right)$ **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Side of square park $a = 50$ m.



Step 1: Area of Square Park [1 Mark]

$$\text{Area of square} = a^2 = 50 \times 50 = 2500 \text{ m}^2$$

Step 2: Area of Four Quadrants and Central Fountain [1 Mark]

Four corner quadrants of radius $r_1 = 7$ m:

$$\text{Area of 4 quadrants} = 4 \times \left(\frac{1}{4}\pi r_1^2\right) = \frac{22}{7} \times 7^2 = 154 \text{ m}^2$$

Central circular fountain of diameter 14 m $\implies r_2 = 7$ m:

$$\text{Area of fountain} = \pi r_2^2 = \frac{22}{7} \times 7^2 = 154 \text{ m}^2$$

Step 3: Remaining Area of Park [1 Mark]

$$\text{Total Developed Area} = 154 + 154 = 308 \text{ m}^2$$

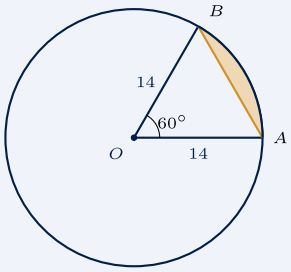
$$\text{Remaining Area} = 2500 - 308 = 2192 \text{ m}^2$$

$$\boxed{\text{Remaining Area of the Park} = \mathbf{2192} \text{ m}^2}$$

Q10. A chord AB of a circle of radius 14 cm subtends an angle of 60° at the centre. Calculate the area of the minor segment. $\left(\text{Use } \pi = \frac{22}{7}, \sqrt{3} = 1.732\right)$ **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Given: Radius $r = 14$ cm, Central Angle $\theta = 60°$.



Step 1: Area of Minor Sector [1 Mark]

$$\text{Area of sector} = \frac{60°}{360°} \times \frac{22}{7} \times 14 \times 14 = \frac{1}{6} \times 616 = \frac{308}{3} \approx 102.67 \text{ cm}^2$$

Step 2: Area of Equilateral $\triangle OAB$ [1 Mark]

Since $OA = OB = 14$ cm and $\angle AOB = 60°$, $\triangle OAB$ is equilateral:

$$\text{Area}(\triangle OAB) = \frac{\sqrt{3}}{4} \times (14)^2 = 49\sqrt{3} = 49 \times 1.732 = 84.868 \text{ cm}^2$$

Step 3: Area of Minor Segment [1 Mark]

$$\text{Area of segment} = \text{Area of sector} - \text{Area}(\triangle OAB)$$

$$= \frac{308}{3} - 84.868 \approx 102.667 - 84.868 = 17.799 \text{ cm}^2 \approx 17.80 \text{ cm}^2$$

$$\boxed{\text{Area of Minor Segment} = \mathbf{17.80} \text{ cm}^2 \quad \left[\text{or } \left(\frac{\mathbf{308}}{\mathbf{3}} - \mathbf{49\sqrt{3}}\right) \text{cm}^2\right]}$$

Q11. A round table cover has six equal designs. If the radius of the cover is 28 cm:

(a) Prove that each design is a minor segment of a circle of central angle 60° and write the formula for its area. **[1.5]**

(b) Calculate the total area of the six designs. $\left(\text{Use } \sqrt{3} = 1.7, \pi = \frac{22}{7}\right)$ **[2]**

(c) Find the total cost of making the designs at the rate of Rs. 0.35 per cm^2. **[1.5]**

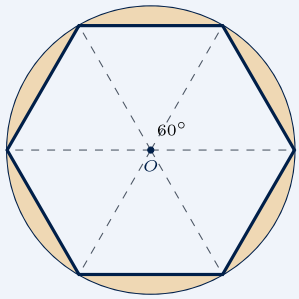
▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Given: Radius $r = 28$ cm, Number of equal designs $n = 6$.

Part (a): Central Angle and Formula Proof [1.5 Marks]

The complete angle at centre is 360°. For 6 equal designs:

$$\theta = \frac{360^\circ}{6} = 60^\circ$$



Each design is bounded by an arc and chord connecting endpoints of two radii, which defines a **minor segment**.

$$\text{Area of 1 design} = \text{Area of sector} - \text{Area}(\triangle OAB) = \frac{60^\circ}{360^\circ}\pi r^2 - \frac{\sqrt{3}}{4}r^2$$

Part (b): Total Area of the Six Designs [2 Marks]

$$\text{Area of 1 sector} = \frac{1}{6} \times \frac{22}{7} \times 28 \times 28 = \frac{1232}{3} \text{ cm}^2$$

$$\text{Area of 1 } \triangle = \frac{\sqrt{3}}{4} \times (28)^2 = 196\sqrt{3} = 196 \times 1.7 = 333.2 \text{ cm}^2$$

$$\text{Total Area} = 6 \times \left(\frac{1232}{3} - 333.2\right) = 2464 - 1999.2 = 464.8 \text{ cm}^2$$

Part (c): Total Cost Calculation [1.5 Marks]

$$\text{Total Cost} = 464.8 \times \text{Rs. } 0.35 = \text{Rs. } 162.68$$

(a) $\theta = \mathbf{60^\circ}$, (b) Total Area = $\mathbf{464.8}$ cm^2, (c) Total Cost = Rs. $\mathbf{162.68}$

Q12. An athletic running track consists of two parallel straightaways and two semicircular ends. The length of each straight segment is 106 m, the distance between the two inner parallel lines is 60 m, and the track is 10 m wide everywhere.

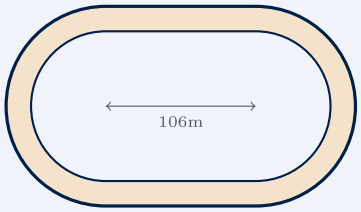
(a) Find the inner perimeter of the track. **[1.5]**

(b) Find the outer perimeter of the track. **[1.5]**

(c) Calculate the total surface area of the running track. $\left(\text{Use } \pi = \frac{22}{7}\right)$ **[2]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]

Given: Straight length $L = 106$ m. Inner diameter $= 60$ m $\implies r = 30$ m. Track width $w = 10$ m $\implies R = 40$ m.



Part (a): Inner Perimeter of the Track [1.5 Marks]

$$\text{Inner Perimeter} = 2L + 2\pi r = 2(106) + 2 \times \frac{22}{7} \times 30$$

$$= 212 + \frac{1320}{7} = \frac{1484 + 1320}{7} = \frac{2804}{7} \text{ m} \approx 400.57 \text{ m}$$

Part (b): Outer Perimeter of the Track [1.5 Marks]

$$\text{Outer Perimeter} = 2L + 2\pi R = 2(106) + 2 \times \frac{22}{7} \times 40$$

$$= 212 + \frac{1760}{7} = \frac{1484 + 1760}{7} = \frac{3244}{7} \text{ m} \approx 463.43 \text{ m}$$

Part (c): Total Surface Area of the Track [2 Marks]

$$\text{Area of 2 straight rectangular strips} = 2 \times (106 \times 10) = 2120 \text{ m}^2$$

$$\text{Area of 2 semicircular ends} = \pi(R^2 - r^2) = \frac{22}{7} \times (40^2 - 30^2) = \frac{22}{7} \times 700 = 2200 \text{ m}^2$$

$$\text{Total Surface Area} = 2120 + 2200 = 4320 \text{ m}^2$$

(a) Inner: $\mathbf{\frac{2804}{7}}$ m $\approx$ **400.57** m, (b) Outer: $\mathbf{\frac{3244}{7}}$ m $\approx$ **463.43** m, (c) Area = **4320** $\mathbf{m^2}$

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Q13. Case Study: Athletic Stadium Running Track Engineering

A sports stadium features an international regulation running track comprising two parallel straight sections of 106 m each, connected by two semicircular ends. The distance between the inner parallel tracks is 60 m, and the track has a uniform width of 10 m.

(i) Find the radius of the inner semicircular boundary of the track. [1]

(ii) Calculate the total distance a runner covers in one full lap along the inner edge. [1]

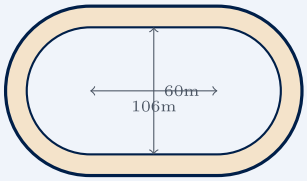
(iii) Calculate the total surface area of the track. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [2]

— OR —

Find the total distance covered in one lap along the outer edge. [2]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 4 Marks]

Given: Straight segment length $L = 106$ m, Inner distance $d = 60$ m, Width $w = 10$ m.



Part (i): Inner Radius of the Track [1 Mark]

$$r = \frac{d}{2} = \frac{60}{2} = 30 \text{ m}$$

Part (ii): Distance Covered in One Full Lap along Inner Edge [1 Mark]

$$\text{Distance}_{\text{inner}} = 2L + 2\pi r = 2(106) + 2 \times \frac{22}{7} \times 30$$

$$= 212 + \frac{1320}{7} = \frac{2804}{7} \text{ m} \approx 400.57 \text{ m}$$

Part (iii): Total Surface Area of the Track [2 Marks]

$$\text{Outer radius } R = 30 + 10 = 40 \text{ m}$$

$$\text{Area} = 2(L \times w) + \pi(R^2 - r^2) = 2(106 \times 10) + \frac{22}{7} \times (1600 - 900)$$

$$= 2120 + 2200 = 4320 \text{ m}^2$$

— OR (Alternative Choice) —

Distance along Outer Edge for One Full Lap: [2 Marks]

$$\text{Distance}_{\text{outer}} = 2L + 2\pi R = 2(106) + 2 \times \frac{22}{7} \times 40$$

$$= 212 + \frac{1760}{7} = \frac{3244}{7} \text{ m} \approx 463.43 \text{ m}$$

(i) $r = \mathbf{30}$ m, (ii) Inner Lap $= \mathbf{\frac{2804}{7}}$ m $\approx \mathbf{400.57}$ m, (iii) Area $= \mathbf{4320}$ m^2 $\left(\text{OR Outer Lap} = \mathbf{\frac{3244}{7}} \text{ m} \approx\right.$

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